

Companion II: Lane A Sharp-Local Construction

Amos Jay Maley

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This companion gives the source-backed detailed proofs for the Lane A sharp-local construction cited by the core paper. It is assembled directly from the canonical March 16, 2026 LaTeX source tree and houses the flowed continuum state, the exact-dimension quotient machinery, the one-shell transport and finite-truncation inverse results, the finite-cap mixed-correlator bridge, the finite-cap sharp-local extension, the inductive-union passage, bounded-base cyclicity in the reconstructed sharp-local Hilbert space, and a theorem-level summary of the state-construction chain. Appendices A–E carry native REVTeX transcriptions of the source-manuscript blocks used for these results. The companion is uniform throughout in native REVTeX form.

GUIDE TO THE DETAILED ARGUMENTS

This source-backed companion is organized to support theorem-level verification while preserving direct access to the March 16, 2026 canonical source tree. The body names the exact core-paper theorems housed here and records where they are proved. Appendices A–E provide native REVTeX transcriptions of the source-manuscript blocks used for the flowed-state opening, the main Lane A construction file, the supporting analytic files, the finite-truncation / mixed-channel closure file, and the normalization / quotient-interface file.

Reference orientation. Standard background used across the theorem-level presentation includes Osterwalder–Schrader reconstruction and lattice reflection positivity^{1–3}, Wilson lattice gauge theory and asymptotic freedom^{4–6}, gradient/Wilson flow and heat-kernel tools^{7–9}, cluster expansion and Dobrushin uniqueness^{10–12}, Bessel-ratio and Toeplitz/total-positivity tools used in the one-plaquette comparison arguments^{13–22}, lattice improvement and weak-coupling comparison inputs^{23–25}, the Yang–Mills problem statement²⁶, DHR superselection theory and local quantum physics^{27–31}, and standard real-algebraic geometry background used in auxiliary reductions³².

Cross-reference to detailed proofs.

Core	Src.	Pages	Detailed proof	First core use
5.1	4.92; 4.93	54–67	Appendix A (+ Companion I, Sec. III for flowed de-patching)	Core Theorem 5.8 and then Core Theorem 5.10
5.2	4.94	67	Appendix A	Core Theorem 5.8 and then Core Theorem VI.1
VI seam	4.95	67	Appendix A	Core Theorem VI.1
5.3	L.8	305	Appendix E	Core Theorem 5.4 and then Core Theorem 5.5
5.4	L.11	307	Appendix E	Core Theorem 5.5 and then Core Theorem 5.6
5.5	5.39	81	Appendix B	Core Theorem 5.6
5.6	5.69	98	Appendix B	Core Corollary 5.7 and then Core Theorem 5.9
5.7	F.245	234	Appendix D	Core Theorem 5.9
5.8	F.330A	278	Appendix D	Core Theorem 5.9
5.9	F.331	280	Appendix D	Core Theorem 5.10
VI bridge	5.74A	103 (in- sert.)	Appendix B	Core Theorem 5.10
5.10	5.74	103	Appendix B	Core Theorem VI.2
VI comp.	5.75	106	Appendix B	Core Theorem VI.2
5.11	5.76	106	Appendix B	Core Theorem VI.2 and then Core Theorem VII.1
5.13	5.77	107	Appendix B	Core Corollary VI.4 and then Core Theorem VII.1

The body sections below are intentionally short. Their role is to identify what each

theorem component establishes, state where the argument is given, and keep the terminology used by the core paper stable. In particular, core Theorem 5.9 is the sole mixed-correlator closure node at finite cap; the promoted finite-cap bridge into Source Theorem 5.74 is the only theorem-level positivity/unitarity input used there; core Corollary 5.11 is the inductive-union passage; and core Theorem 5.13 is the first bounded-base cyclicity export. Section 7 later assembles the same functional backbone into one local theorem-level chain. This companion keeps that organization explicit.

I. FLOWED CONTINUUM STATE AND BOUNDED POSITIVE-TIME BASE

This section houses the entry point of Lane A used by core Section V. In the March 16, 2026 source manuscript the corresponding theorem-level results are Theorem 4.92 and Corollaries 4.93–4.95. Together they certify the unique flowed continuum state, the executive Wilson/RG identification on A_{flow} , the tuned bounded positive-time base state, and the dyadic reflected matrix-element limits later consumed on the QE3 seam.

Detailed proof. The full source file is transcribed natively in Appendix A; the flowed de-patching seam used for the Wilson/RG identification on A_{flow} is proved in Companion I, Section III..

The role of the present section is front-end only. It certifies the starting state for Lane A and surfaces the Component 4 dyadic OS-limit seam, but it does not yet provide exact-dimension quotient control, finite-cap closure, or any continuum transport theorem.

II. EXACT-DIMENSION QUOTIENT BLOCKS AND COEFFICIENT EXTRACTION

This section houses the quotient-block theorems that underwrite the canonical exact-dimension formalism. In the March 16, 2026 source manuscript the corresponding results are Theorem L.8 and Theorem L.11 in Appendix L. Their theorem-level role is to identify the exact-dimension quotient action of the small-flow substitution and to extract the canonical coefficient data on arbitrary quotient blocks before any finite-cap closure theorem is invoked.

Detailed proof. The detailed exact-dimension proof transcriptions are given natively in

Appendix E..

The body of the core paper uses these results only through their theorem-level content: the exact-dimension quotient results are the canonical input later consumed by the one-shell transport and finite-truncation inverse file.

III. ONE-SHELL TRANSPORT AND FINITE-TRUNCATION INVERSE CONTROL

This section houses the finite-truncation results used throughout core Section V. In the March 16, 2026 source manuscript the corresponding results are Theorem 5.39, Theorem 5.69, and Corollary F.245. Together they provide the one-shell transport on finite truncations, the unconditional finite-truncation inverse control, and the accompanying SFTE/remainder package.

Detailed proof. The detailed proof transcriptions for Theorems 5.39, 5.69, 5.74A, 5.74, 5.75, 5.76, and 5.77 are given natively in Appendix B. The proof of Corollary F.245 is transcribed natively in Appendix D. Supporting flow-localization, kernel, and insertion-support material is recorded in Appendix C..

This section should be read as the point where the old sample-sector dependence is discharged at theorem level. The finite-truncation inverse statement is unconditional in the proof architecture carried by the core paper; the detailed coefficient and insertion-RG file is carried in the supporting appendices of this companion.

IV. FINITE-CAP MIXED-CORRELATOR BRIDGE

This section houses the mixed-channel finite-cap seam used by the core paper. In the March 16, 2026 source manuscript the corresponding results are Corollary F.330A and Theorem F.331. Their theorem-level role is sharply separated: Corollary F.330A packages the finite mixed bounded-family input, Theorem F.331 is the sole mixed-correlator closure node at finite cap, and the promoted Source Theorem 5.74A in Appendix B packages the theorem-level positive/unital finite-cap state input consumed next by Source Theorem 5.74.

Detailed proof. The theorem-level proof transcription is given in Appendix D..

This role split should be kept literal. The finite mixed-family packaging is not yet the closure theorem, and no inductive-union or cyclicity statement should be read into this section. Section 7 records the resulting state-construction chain in one theorem-level summary.

V. FINITE-CAP SHARP-LOCAL EXTENSION

This section houses the capwise extension theorem cited by core Section V. In the March 16, 2026 source manuscript the corresponding theorem-level results are the promoted bridge Source Theorem 5.74A, Source Theorem 5.74, and the immediate bounded-state restriction corollary 5.75. The theorem-level function of Source Theorem 5.74A is to expose the positive/unital finite-cap bridge from the bounded-family and mixed-channel closure results; Source Theorem 5.74 then extends the flowed Lane C state to the finite-cap sharp-local setting and records the capwise OS structure; Corollary 5.75 records only the bounded-base restriction compatibility later consumed on the QE3 seam.

Detailed proof. The detailed proof transcription is given in Appendix B. Supporting flow-localization, kernel, and insertion estimates are recorded in Appendix C..

The finite-cap extension is the first theorem on the Lane A line that actually reaches the sharp-local capwise algebra. Its bounded-state restriction corollary is a separate compatibility node and remains distinct from the later inductive-union and cyclicity exports. Appendix B now also isolates the finite-cap positivity mechanism in a dedicated theorem-level lemma cited directly by Source Theorem 5.74, and it writes the overlap comparison used later by Source Corollary 5.76 directly on shared generators and inverse-flow representatives.

VI. INDUCTIVE-UNION PASSAGE AND BOUNDED-BASE CYCLICITY

This section houses the final two Lane A exports used by the core paper. In the March 16, 2026 source manuscript the corresponding results are Corollary 5.76 and Theorem 5.77, following the earlier bounded-state restriction compatibility corollary 5.75. Their theorem-level roles are again distinct: Corollary 5.76 passes from finite caps to the full sharp-local inductive union, while Theorem 5.77 is the first bounded-base cyclicity statement on the reconstructed sharp-local Hilbert space. The seam language is now frozen so that no reader can mistake Corollary 5.75 for the inductive-union theorem or mistake Source

Theorem 5.74(b) for a full-scope statement.

Detailed proof. The detailed proof transcriptions are given in Appendix B..

Within the architecture of the core paper, these two results must remain separate. Core Corollary 5.11 is the inductive-union passage; core Theorem 5.13 is the bounded-base cyclicity export. Appendix B now makes both seam steps theorem-local: the inductive-union proof includes the explicit overlap comparison, and the cyclicity proof includes the explicit bounded approximating sequence and telescoping estimate. Neither should be blurred into the earlier mixed-channel seam.

VII. STATE-CONSTRUCTION SUMMARY

This section packages the functional and measure backbone of Lane A in one place. Its role is theorem-level: it records exactly where the state comes from, where finite-cap positivity is proved, and where the passage to the full sharp-local Hilbert space occurs.

Proposition VII.1 (State-construction summary for the Lane A functional backbone). *Fix a weak-window target value $u_{\text{ren}} \in (0, u_*]$. The state-construction chain used later by the QE3 seam and the endpoint theorems is exactly the following.*

- (i) *Source Corollary 4.94 supplies the bounded Wilson reflection-positive base state $\omega_{\text{bd}}^{(u_{\text{ren}})}$ on A_+ , together with the dyadic positive-time matrix elements later used on the QE3 seam.*
- (ii) *For every fixed engineering cap and every finite bounded family $W \subset *(A_{\text{flow}}^{(\leq \Delta_{\text{max}})}, A_+)$, Source Corollary F.330A packages the joint Wilson limits into compatible positive unital capwise states on $\text{Alg}(W)$.*
- (iii) *For every fixed engineering cap and fixed mixed channel, Source Theorem F.331 closes mixed correlators against fixed capwise words, while Source Corollary 5.70 supplies the finite-truncation inverse/remainder results used to identify the formal centered local symbols with the corresponding renormalized local insertions.*
- (iv) *Source Theorem 5.74A upgrades those mixed-channel limits to a unique positive unital capwise functional on the abstract finite-cap sharp-local algebra, and it isolates the*

positivity mechanism theorem-locally: reflection positivity is read only through bounded positive-time representatives, and the inverse-flow replacement error tends to zero.

- (v) *Source Theorem 5.74 turns that capwise functional into the finite-cap sharp-local state with capwise OS structure and finite-cap field realization on the common invariant core.*
- (vi) *Source Corollary 5.76 glues the compatible finite-cap states to the unique inductive-union sharp-local state*

$$A_{\text{loc}} = \varinjlim_{\Delta_{\text{max}} \rightarrow \infty} A_{\text{loc}}^{(\leq \Delta_{\text{max}})},$$

which extends the bounded base state on A_+ by an explicit overlap comparison on shared generators and inverse-flow representatives.

- (vii) *Source Theorem 5.77 is the first full-sharp-local density export: the image of A_+ is dense in the reconstructed Hilbert space and the closed linear span of the centered bounded classes is the vacuum-orthogonal sector; the proof now gives the explicit bounded approximating sequence used later at QE3.*

No later QE3 or endpoint theorem uses any additional state-construction or positivity step beyond this seven-node chain.

Detailed proof. Items (i)–(vii) are proved respectively in Appendix A (Source Corollary 4.94), Appendix D (Source Corollary F.330A and Source Theorem F.331), and Appendix B (Source Theorem 5.74A, Source Theorem 5.74, Source Corollary 5.76, and Source Theorem 5.77)..

Dependency summary.

Inputs. Source Corollary 4.94, Source Corollary F.330A, Source Theorem F.331, Source Corollary 5.70, Source Theorem 5.74A, Source Theorem 5.74, Source Corollary 5.76, and Source Theorem 5.77

Imported standard background. none beyond the standard OS reconstruction step already isolated where the finite-cap and full sharp-local states are constructed

Conclusion at this node. the live Lane A functional backbone is exactly a seven-step chain from the bounded Wilson base state to the full sharp-local state and its bounded-base density export; no hidden positivity or measure-transport step remains outside this summary

Used later in. the QE3 seam in Companion I / core Section VI and the endpoint theorems in Companion III / core Section VII

Remark VII.2 (What the summary answers immediately). For the construction of the state, read items (i)–(vi). For positivity, read item (iv) together with the dedicated finite-cap positivity lemma cited there and in Source Theorem 5.74. Nothing later in the manuscript requires a different functional backbone.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Appendix A: Native REVTeX transcription of current Section 4 source block (pp. 27–67)

This appendix gives a native theorem-level transcription of the Section 4 source block from the March 16, 2026 manuscript. Its focus is the flowed continuum state and the dyadic-RG / no-subsequence convergence infrastructure culminating in source Theorem 4.92 and Corollaries 4.93–4.95.

1. Group-uniform A1 input and the dyadic GF step-scaling lane

The source Section 4 begins by isolating the one-plaquette coefficient input and the dyadic gradient-flow step-scaling structure used throughout Lane C.

Theorem A.1 (Source Theorem 4.2: group-uniform A1 input used on the primary spine). *For every compact connected simple gauge group G and every fixed finite-dimensional unitary Wilson representation ρ , the one-plaquette coefficients satisfy the crossover bound of the compact-simple A1 theorem. Equivalently, on the primary Route 1 spine every use of the A1 gate is available with group-dependent constants $(c_{A1}(G, \rho), \beta_{A1}(G, \rho))$ and no remaining Lie-type restriction.*

Proposition A.2 (Source Proposition 4.6: one-loop coefficient is scheme invariant). *Let u and u' be two renormalized couplings related near 0 by an analytic reparametrization*

$$u' = u + a_1 u^2 + a_2 u^3 + O(u^4).$$

If

$$\beta(u) = -2b_0u^2 + O(u^3),$$

then

$$\beta'(u') = -2b_0(u')^2 + O((u')^3).$$

In particular, the coefficient b_0 is invariant under the reparametrization.

Lemma A.3 (Source Lemma 4.7: beta-remainder implies step-scaling remainder). *Assume there exist constants $u_0 > 0$ and $B \geq 0$ such that*

$$|\beta_{\text{GF}}(u) + 2b_0u^2| \leq Bu^3 \quad (0 \leq u \leq u_0),$$

and assume u_0 is small enough that $D := 1 - 2b_0u_0 \log 2 > 0$. Then for every $u \in [0, u_0]$, the factor-2 step-scaling map satisfies

$$\Sigma(u) = u + 2b_0u^2 \log 2 + R_\Sigma(u), \quad |R_\Sigma(u)| \leq C_\Sigma u^3$$

with explicit constant

$$C_\Sigma := \frac{(2b_0 \log 2)^2}{D} + \frac{B \log 2}{D^3}.$$

2. One-step analytic RG map and dyadic beta remainder

The middle of the source Section 4 turns the Lane C dyadic RG into a scale-uniform analytic map with explicit Taylor remainder bounds.

Proposition A.4 (Source Proposition 4.9: analytic RG map with explicit Taylor remainder bounds). *There exist constants*

$$r_0 > 0, \quad \kappa \in (0, 1), \quad A_2 \in \mathbb{R}, \quad C_u > 0, \quad C_K > 0,$$

and linear operators $L_j : \mathcal{K}_j \rightarrow \mathcal{K}_{j+1}$ with $\|L_j\|_{\text{op}} \leq \kappa$, such that for all j and all $(u, K) \in B_j(r_0)$ the RG step satisfies

$$u' = u + A_2u^2 + \delta_u(u, K), \tag{A1}$$

$$K' = L_j K + \delta_K(u, K), \tag{A2}$$

with bounds

$$|\delta_u(u, K)| \leq C_u(|u|^3 + |u| \|K\|_j), \tag{A3}$$

$$\|\delta_K(u, K)\|_{j+1} \leq C_K(|u|^2 + \|K\|_j^2). \quad (\text{A4})$$

All constants are uniform in the dyadic scale.

Proposition A.5 (Source Proposition 4.10: entrance estimate). *There exists an entrance scale j_* and a constant $D > 0$ such that*

$$\|K_{j_*}\|_{j_*} \leq Du_{j_*}^2, \quad |u_{j_*}| \leq r_0/2.$$

Proposition A.6 (Source Proposition 4.11: quadratic coefficient identification). *The RG marginal coefficient satisfies*

$$A_2 = 2b_0 \log 2, \quad b_0 > 0$$

for pure Yang–Mills. In the proof only the positivity $b_0 > 0$ is used downstream.

Lemma A.7 (Source Lemma 4.13: irrelevant sector stays $O(u^2)$). *Assume the analytic RG map above. Define the cone region*

$$\mathcal{C}_j(M) := \{(u, K) \in B_j(r_0) : \|K\|_j \leq Mu^2\}.$$

If

$$M \geq \frac{2C_K}{1 - \kappa}$$

and r_0 is sufficiently small, then

$$(u, K) \in \mathcal{C}_j(M) \implies (u', K') \in \mathcal{C}_{j+1}(M).$$

Corollary A.8 (Source Corollary 4.14: uniform irrelevant bound along the dyadic trajectory). *With*

$$M := \max\left\{D, \frac{2C_K}{1 - \kappa}\right\},$$

one has for all $j \geq j_*$,

$$\|K_j\|_j \leq Mu_j^2$$

provided the RG stays in the small-coupling ball.

Theorem A.9 (Source Theorem 4.15: dyadic step remainder bound in RG seminorm topology). *Assume the preceding RG hypotheses. Then for all $j \geq j_*$ and all sufficiently small u_j ,*

$$|u_{j+1} - u_j - A_2 u_j^2| \leq C_\Sigma u_j^3, \quad C_\Sigma := C_u(1 + M).$$

The constant C_Σ is uniform in j and hence uniform in a/L_j .

Corollary A.10 (Source Corollary 4.16: uniform dyadic beta-remainder bound). *Define the dyadic beta function by*

$$\beta^{(2)}(u_j) := -\frac{u_{j+1} - u_j}{\log 2}.$$

Then for all $j \geq j_$ and all sufficiently small u_j ,*

$$|\beta^{(2)}(u_j) + 2b_0 u_j^2| \leq B u_j^3, \quad B := \frac{C_\Sigma}{\log 2} = \frac{C_u(1+M)}{\log 2},$$

uniformly in the dyadic scale.

3. Uniform convexity and the extracted-irrelevant scaling gap

The source Section 4 next reduces the one-step fluctuation estimates to finite-dimensional uniform convexity on the good-chart domain together with the canonical extracted-irrelevant scaling gap.

Lemma A.11 (Source Lemma 4.40: explicit lower bound for the tree-gauge curl coercivity). *For the $2 \times 2 \times 2 \times 2$ coarse block with the canonical tree gauge, the reduced integer matrix*

$$Q := D_{\text{red}}^* D_{\text{red}}$$

has smallest eigenvalue λ_ satisfying*

$$\lambda_* > \frac{1}{4}.$$

Consequently the same lower bound holds for the full G operator $D_{\text{red}}^ D_{\text{red}}$, since it is block-diagonal over Lie components.*

Theorem A.12 (Source Theorem 4.44: uniform convexity on the chart domain). *Fix $\delta \leq \min\{\pi/4, \delta_{\text{Haar}}(G)\}$ and define*

$$r_* := \sup_{\|A\|_\infty \leq \delta} \max_{p \in P(Q)} \|F_p(A)\|.$$

Assume the BCH derivative bounds and the smallness conditions

$$\varepsilon_1(\delta) \leq \frac{1}{2} \sqrt{\lambda_*}, \quad M(r_*; G) \varepsilon_2(\delta) r_* \leq \frac{\alpha_{\min}}{8} m(r_*; G) \lambda_*.$$

Then for all A with $\|A_e\| \leq \delta$,

$$\nabla^2 H_{j+1}(A) \succeq \varrho I, \quad \varrho := \alpha_{\min} \beta \frac{m(r_*; G) \lambda_*}{8}.$$

In particular, $\varrho > 0$ is uniform in scale j provided $\beta \geq \beta_{\min} > 0$ and δ is fixed.

Lemma A.13 (Source Lemma 4.46: scaling gap after extraction). *Assume the extraction removes all gauge-invariant local components of engineering dimension ≤ 4 . Then every remaining local monomial has dimension at least 6, so the minimal dyadic scaling gap is*

$$\Delta_{\min} = 6 - 4 = 2.$$

Under deterministic dyadic rescaling one therefore has

$$\|\text{Rescale}(K)\|_{j+1} \leq 2^{-\Delta_{\min}} \|K\|_j = \frac{1}{4} \|K\|_j$$

for extracted irrelevants, provided the seminorm is defined with canonical scaling weights.

Theorem A.14 (Source Theorem 4.66: uniform dyadic beta-remainder bound for the patched RG). *Assume:*

1. *the dyadic RG step is the chart-truncated map R_j^{good} ,*
2. *extraction removes all relevant/marginal components up to the chosen depth,*
3. *the identification $A_2 = 2b_0 \log 2$ holds with $b_0 > 0$,*
4. *the project seminorm axioms hold with scale-independent constants.*

Then there exist $u_0, B > 0$, independent of the dyadic scale j , such that for all scales in the asymptotically free window and all $u \in [0, u_0]$,

$$|\beta^{(2)}(u) + 2b_0 u^2| \leq B u^3,$$

with explicit choice

$$B = \frac{C_u(1+M)}{\log 2}, \quad M := \max\left\{D, \frac{2C_K}{1-\kappa}\right\}.$$

4. Stable manifold, injective tuning, and observable Lipschitz control

The last large block of the source Section 4 constructs the UV-stable irrelevant trajectory, proves injective tuning of the renormalization map at fixed physical scale, and establishes the observation-scale Lipschitz dependence needed for uniqueness of the continuum state.

Theorem A.15 (Source Theorem 4.72: stable manifold for the patched RG). *Let $(u_j)_{j \leq 0}$ be an admissible asymptotically free trajectory. Then:*

1. for each fixed $j \leq 0$, the limit

$$K_j^{(\infty)} := \lim_{N \rightarrow \infty, N \geq |j|} K_j^{(N)}$$

exists in $\mathcal{K}_j$ and satisfies the RG recursion

$$K_{j+1}^{(\infty)} = \mathcal{K}_j(u_j, K_j^{(\infty)});$$

2. one has $K_j^{(\infty)} \rightarrow 0$ in norm as $j \rightarrow -\infty$;

3. the UV-stable irrelevant trajectory is unique;

4. the limit is universal with respect to bounded UV initialization in the irrelevant sector.

Moreover,

$$\|K_j^{(\infty)}\|_j \leq C_F \sum_{m=-\infty}^{j-1} \kappa^{j-1-m} u_m^2.$$

Theorem A.16 (Source Theorem 4.78: injectivity/monotonicity of the renormalization map). *Assume the one-step analytic remainder bounds for the step-scaling maps. Then there exists a uniform asymptotically free interval $[0, u^\circ]$ such that for every cutoff depth N the renormalization map*

$$\rho_N(u_{-N}) := u_0$$

is C^1 and strictly increasing on $[0, u^\circ]$. In particular, for every target u_μ in the image of ρ_N there exists a unique tuned UV coordinate u_{-N} with $\rho_N(u_{-N}) = u_\mu$.

Theorem A.17 (Source Theorem 4.86: bulk observable Lipschitz in the RG data). *Fix a finite observable set $\mathcal{F} \subset A_{\text{flow}}$ and choose the observation scale j_{obs} attached to $\mathcal{F}$. Then for any two datasets $(u, K), (\tilde{u}, \tilde{K})$ in the observation-scale RG ball,*

$$\max_{F \in \mathcal{F}} |G_F(u, K) - G_F(\tilde{u}, \tilde{K})| \leq L_{\mathcal{F}} (|u - \tilde{u}| + \|K - \tilde{K}\|_{j_{\text{obs}}}),$$

with explicit constant $L_{\mathcal{F}}$ independent of the lattice spacing in the patched construction.

5. No-subsequence convergence and the flowed-state theorem

The source Section 4 ends by assembling the no-subsequence convergence theorem for the patched outputs and its identification with the physical Wilson continuum state.

Theorem A.18 (Source Theorem 4.92: no-subsequence convergence for the flowed continuum state). *Assume uniform boundedness/precompactness, injective tuning, stable-manifold uniqueness, and observation-scale Lipschitz control. Fix a target renormalized coupling value in the asymptotically free interval, and for each lattice spacing tune the UV coupling so that the renormalization condition holds. Then:*

1. *for every finite set $\mathcal{F} \subset A_{\text{flow}}$, the joint values $\{\omega_a^{\text{RG}}(F)\}_{F \in \mathcal{F}}$ converge as $a \rightarrow 0$ without subsequence extraction;*
2. *there exists a unique state ω on A_{flow} such that*

$$\omega(F) = \lim_{a \rightarrow 0} \omega_a(F) = \lim_{a \rightarrow 0} \omega_a^{\text{RG}}(F) \quad (F \in A_{\text{flow}});$$

3. *the limit state is Euclidean invariant and reflection invariant, and inherits the algebraic identities preserved under pointwise limits.*

Corollary A.19 (Source Corollary 4.93: unique continuum Euclidean state on A_{flow}). *Under the hypotheses of Theorem 4.92, the tuned unpatched Wilson expectations ω_a converge (without subsequences) to a unique Euclidean-invariant state ω on A_{flow} for all flow times $t \in (0, t_*]$.*

Proof. This is immediate from Theorem 4.92, parts (ii)–(iii). □

Corollary A.20 (Source Corollary 4.94: tuned dyadic Wilson limit on the bounded positive-time algebra). *Fix a target value $u_{\text{ren}} \in (0, u_*]$ and the associated tuned dyadic trajectory*

$$a_n = \ell_0 2^{-n}, \quad \beta_n = \beta(a_n; u_{\text{ren}}).$$

Then for every finite family $\mathcal{F} \subset A_+$ of bounded cylinder observables supported in $\{x_0 > 0\}$, the Wilson expectations

$$\{\omega_{a_n}(F)\}_{F \in \mathcal{F}}$$

converge jointly as $n \rightarrow \infty$. In particular, for every $F \in A_+$ the limit $\lim_{n \rightarrow \infty} \omega_{a_n}(F)$ exists and defines a state on A_+ .

Corollary A.21 (Source Corollary 4.95: tuned dyadic convergence of reflected matrix elements). *Fix the same tuned dyadic trajectory and define the dense dyadic translation set*

$$D_{\text{dyad}} := \{m \ell_0 2^{-k} : m, k \in \mathbb{N}_0\}.$$

Then for every finite family built from finitely many elements of A_+ by applying finitely many reflections and finitely many translations τ_s with $s \in D_{\text{dyad}}$, the Wilson expectations converge jointly along the tuned dyadic sequence. In particular, for every bounded $F, G \in A_+$ and every $s \in D_{\text{dyad}}$, the limit

$$\lim_{n \rightarrow \infty} \omega_{a_n}(\Theta F \tau_s G)$$

exists.

Remark A.22. For the purposes of the core paper, the theorem-level outputs of the present appendix are source Theorem 4.92 and Corollaries 4.93–4.95. Corollary 4.93 records the executive flowed-state formulation on A_{flow} , Corollary 4.94 gives the tuned bounded positive-time base state, and Corollary 4.95 gives the dyadic reflected matrix-element limits later consumed at the QE3 seam. The Wilson/RG identification on A_{flow} is paired in the live suite with the flowed de-patching seam proved in Companion I, Section III. The larger dyadic-RG and injective-tuning machinery is retained here because it is part of the same native source block and because later Lane A closure theorems continue to consume its outputs.

Appendix B: Native transcription of the main Lane A construction file (source pp. 68–108)

This appendix replaces the principal Lane A construction facsimile block with a native REVTeX presentation of the March 16, 2026 source-manuscript Section 5 file. Its role in the present companion is theorem-level: it records the quasi-local flowed probes, the insertion-RG and OS-seminorm transfer package, the canonical exact-dimension block machinery, the finite-truncation inverse file, the finite-cap mixed-correlator bridge, the finite-cap sharp-local extension, the inductive-union passage, and bounded-base cyclicity in the reconstructed sharp-local Hilbert space. The presentation below follows the source ordering and preserves the source labels in theorem statements. At the load-bearing closure nodes we replace the earlier synopsis-level presentation by explicit proof transcriptions in theorem-level form.

1. Flowed probes, RP-safe representatives, and insertion-RG infrastructure

The source Section 5 opens by correcting the naive finite-propagation intuition for gradient flow. The theorem-level point is not strict support in a radius- $\sqrt{t}$ ball, but quasi-locality with

Gaussian off-diagonal suppression, together with a truncation rule that produces RP-safe positive-time representatives for later use inside the OS pairings.

Theorem B.1 (Source Theorem 5.12: flow quasi-locality and derivative bounds). *Fix $t_* > 0$ and a finite derivative order $r_{\max}$. Under the bounded-regime control of the good-flow chart event and the Gaussian kernel bounds for the linearized DeTurck operator, the following hold for the flowed probe family:*

- (i) *functional derivatives of the flowed scalar and pseudoscalar probes satisfy Gaussian off-diagonal bounds up to order $r_{\max}$;*
- (ii) *for any $\eta > 0$ there exists a radius function $R(t) \asymp \sqrt{\log(1/t)}$ such that modifying the initial data outside the ball of radius $R(t)\sqrt{t}$ changes the flowed probe by at most $O(t^\eta)$ in supremum norm;*
- (iii) *all constants can be chosen uniformly in the lattice spacing for $t \geq ca^2$, provided the chart/boundedness hypothesis is uniform in a .*

Lemma B.2 (Source Lemma 5.72: RP-safe positive-time representatives). *Let f be supported in $\{x_0 \geq \varepsilon\}$ and choose $R \geq 1$ with $R\sqrt{t} \leq \varepsilon/2$. Then each flowed probe admits a bounded cylinder truncation depending only on underlying variables in the strict positive-time algebra A_+ , and replacing the flowed probe by this positive-time representative changes every OS/RP pairing by an amount controlled by the quasi-local tail.*

Lemma B.3 (Source Lemma 5.73: bad-event contribution is negligible in OS/RP pairings). *For the same setup, the difference between a flowed probe and its RP-safe representative is bounded by the sum of a good-event quasi-locality error and a bad-event probability term. The latter is superpolynomially small in the asymptotically free window and can therefore be absorbed into the small-flow-time error budget used later in the Lane A closure file.*

Proof synopsis. The source proof first reduces the flow analysis to the DeTurck-gauge system, then proves Gaussian kernel bounds for the linearized operator and its derivatives, and finally combines those bounds with the local good-flow chart event to obtain truncated positive-time representatives. The theorem-level consequence is that every later use of reflection positivity in Lane A acts only on the unflowed positive-time algebra, even when flowed probes enter as the analytic input..

The source next builds the insertion-RG and the OS-seminorm transfer machinery used throughout the one-shell and finite-cap arguments.

Proposition B.4 (Source Proposition 5.37: one-step insertion integration bound). *On the RG ball, the one-step insertion integration operator satisfies*

$$\|E_{j+1}^{\text{good}}(u_j, K_j)J_j\|_{\text{ins};j+1;k,K,\varepsilon} \leq C_{E,\text{ins}} \|J_j\|_{\text{ins};j;k,K,\varepsilon},$$

with an explicit constant $C_{E,\text{ins}}$ independent of the dyadic scale and independent of the lattice spacing on the chart-truncated regime.

Theorem B.5 (Source Theorem 5.31: insertion RG decomposition at scale μ). *Assume the extracted-irrelevant insertion sector contracts. Then after iterating the patched insertion RG to the matching scale μ , every flowed insertion admits a decomposition into finitely many renormalized local operators plus an extracted-irrelevant remainder whose insertion norm decays like a positive power of $\mu^2 t$.*

Theorem B.6 (Source Theorem 5.45: insertion seminorm implies OS seminorm). *For positive-time insertion activities supported in a fixed compact region,*

$$\|F_J\|_{\text{OS};k,K,\varepsilon} \leq C_{\text{OS} \leftarrow \text{ins}} \|J\|_{\text{ins};j;k,K,\varepsilon},$$

with scale-independent constant $C_{\text{OS} \leftarrow \text{ins}}$ under the patched-chart construction.

These results are the mechanism by which insertion-RG remainder bounds are converted into the OS/core estimates used later in the finite-cap closure theorem.

2. Canonical exact-dimension blocks and finite-truncation inverse control

The middle of source Section 5 replaces the earlier sample-sector dependence by a theorem-level exact-dimension block formalism. The scalar/tensor/loop sectors remain in the long source as illustrations; the load-bearing widening step is the canonical block package and its finite-truncation assembly.

Theorem B.7 (Source Theorem 5.39: canonical block one-shell increment). *Fix an exact-dimension symmetry block $s = (\Delta, \epsilon, \lambda)$ of the quotient formalism and let $Y_{s,j}(f)$ denote the*

canonical flowed lifts at dyadic times t_j . Then there exist scale-independent constants and canonically defined linear maps on the quotient block such that

$$Y_{s,j+1}(f) = (I_s + u_j A_{s,j}) Y_{s,j}(f) + \sum_{s' \prec_s} u_j B_j^{ss'} Y_{s',j}(f) + Q_{s,j}(f),$$

with uniformly bounded block operators and a remainder satisfying

$$\|Q_{s,j}(f)\|_{L^2} \leq C_{s,f} u_j^2.$$

Equivalently, after quotienting by lower blocks, the exact-dimension one-shell transport has the basis-free form

$$S_j^s = I_s + u_j A_{s,j} + R_{s,j}, \quad \|R_{s,j}\|_{\text{op}} \leq C_s u_j^2.$$

Theorem B.8 (Source Theorem 5.69: basis-independent finite-truncation inverse control). *Fix an engineering cutoff $\Delta_{\max} \geq 0$ and order the canonical symmetry blocks of engineering dimension at most $\Delta_{\max}$ by the source grading order. Let $C^{(\leq \Delta_{\max})}(t, \mu)$ be the full coefficient matrix from the canonical flowed lifts of that truncation to the corresponding renormalized local basis. Then for sufficiently small t :*

- (i) *the matrix is block lower triangular in the grading order;*
- (ii) *each diagonal block satisfies the polylogarithmic inverse bound coming from the canonical block UV theorem;*
- (iii) *each off-diagonal block is uniformly bounded on the small-flow window.*

Consequently there exist constants $C_{\Delta_{\max}, \text{inv}}$ and $p_{\Delta_{\max}}$ such that

$$\left\| \left(C^{(\leq \Delta_{\max})}(t, \mu) \right)^{-1} \right\|_{\text{op}} \leq C_{\Delta_{\max}, \text{inv}} \left(\frac{u(\mu)}{u(\mu_t)} \right)^{p_{\Delta_{\max}}}$$

for $0 < t \leq t_0$.

Corollary B.9 (Source Corollary 5.70: Lane A closure on every finite truncation). *For every finite engineering cutoff $\Delta_{\max}$, the corresponding truncation of the canonical sharp-local basis carries both the quantitative inverse-control package and the SFTE/remainder-compatibility package required by the finite-cap Lane A extension theorem.*

Proof of Source Theorem 5.39, Source Theorem 5.69, and Source Corollary 5.70. The exact-dimension diagonal input comes from Appendix E. On each exact-dimension symmetry

block, the associated-graded small-flow map is the identity and the tree-level exact-dimension coefficient-extraction matrix is therefore the identity. The one-shell transport theorem records the quantitative upgrade of that exact-dimension statement: after one dyadic shell step the flowed lift on a fixed block is the identity plus an $O(u_j)$ block operator, lower blocks receive only strictly lower-grade contributions, and the remainder is $O(u_j^2)$ in the capwise L^2 topology.

Once the blocks are ordered by engineering dimension and symmetry type, power counting makes the full finite-truncation coefficient matrix block lower triangular. The exact-dimension block theorem from Appendix E controls the diagonal UV matching blocks, while the one-shell transport relation upgrades those diagonal blocks to the polylogarithmic inverse bounds required at the physical matching scale. For the finitely many off-diagonal blocks in a fixed truncation, the insertion integration bound and the insertion-to-OS transfer theorem supply scale-uniform bounds; finite depth in a fixed truncation only changes the constant. Because the truncation is finite, the lower-triangular assembly produces one global inverse bound

$$\left\| \left(C^{(\leq \Delta_{\max})}(t, \mu) \right)^{-1} \right\|_{\text{op}} \leq C_{\Delta_{\max}, \text{inv}} \left(\frac{u(\mu)}{u(\mu_t)} \right)^{p_{\Delta_{\max}}}$$

for all sufficiently small t .

The corollary is then immediate. The finite-truncation small-flow-time expansion has a block-lower-triangular coefficient matrix with the same inverse bound, and the capwise remainder is $O(t^{\eta_{\Delta_{\max}}})$ in every fixed vacuum or mixed channel. Hence every finite truncation carries both the quantitative inverse-control package and the SFTE / remainder-compatibility package required later by the finite-cap Lane A extension theorem. $\square$

The theorem-level significance of this block is that finite-truncation inverse control is no longer sample-sector-dependent. It is canonical and capwise, and it is the exact input later consumed by the mixed-channel finite-cap closure theorem.

3. Finite-cap mixed-channel closure and sharp-local extension

The source section then enters the finite-cap closure stage. The core paper deliberately keeps the theorem-level separation literal: the mixed bounded-family packaging node, the first mixed-correlator closure node at finite cap, the capwise sharp-local extension, the inductive-union passage, and bounded-base cyclicity are all distinct.

Corollary B.10 (Source Corollary F.330A: finite mixed bounded-family packaging). *At every fixed engineering cap, the mixed bounded-family packaging required for the finite-cap closure theorem is available as a finite family of compatible bounded capwise states.*

Theorem B.11 (Source Theorem F.331: first mixed-correlator closure node at finite cap). *The finite-cap mixed-channel seam closes at theorem level: replacing the designated slot by the bounded inverse-flow representative changes every mixed correlator by $o(1)$ as the flow parameter tends to zero, against every fixed capwise word in the remaining slots. This theorem is the first mixed-correlator closure node on the Lane A chain and the sole mixed-channel closure node cited by the core paper.*

Theorem 5.74A (Finite-cap state bridge into Source Theorem 5.74). *Fix an engineering cutoff $\Delta_{\max}$ and let $A_{\text{loc}}^{(\leq \Delta_{\max})^\wedge}$ denote the abstract unital $*$ -algebra generated by the bounded capwise flowed sector, the bounded positive-time algebra, and the formal centered capwise local symbols. Assume the finite-family packaged bounded capwise states of Source Corollary F.330A, the mixed-correlator closure theorem Source Theorem F.331, and the finite-truncation inverse/remainder package of Source Corollary 5.70. Then:*

- (i) *the inverse-flow limits determine a unique positive unital capwise functional on $A_{\text{loc}}^{(\leq \Delta_{\max})^\wedge}$ extending the capwise flowed state and the tuned bounded positive-time base state on their respective bounded subalgebras;*
- (ii) *reflection positivity is read only through bounded positive-time representatives, and the inverse-flow replacement error tends to zero; equivalently, the corresponding OS pairings are inherited from the tuned bounded Wilson base state through RP-safe representatives in A_+ and remain nonnegative in the capwise limit;*
- (iii) *this bridge provides Source Theorem 5.74 exactly the capwise state-existence, positivity/unitality, and finite-cap OS input package; it does not yet assert the inductive-union passage or bounded-base cyclicity.*

Proof. Source Corollary F.330A packages one finite mixed bounded family at a time into compatible positive unital capwise states. Source Theorem F.331 upgrades those packaged states to the unique capwise mixed-correlator functional and fixes its restrictions to the bounded flowed and bounded positive-time sectors. Source Corollary 5.70 then supplies the quantitative inverse-control and remainder-compatibility data needed to identify the formal

centered symbols with the renormalized local generators used in Source Theorem 5.74. For reflection positivity one replaces the flowed probes inside the inverse-flow representatives by their RP-safe bounded positive-time representatives before evaluating the OS pairing in the tuned bounded Wilson base state; the replacement error is $o(1)$, so nonnegativity passes to the capwise limit. No statement about the inductive union or bounded-base cyclicity is used here. $\square$

Lemma B.12 (Finite-cap positivity mechanism). *Fix an engineering cutoff $\Delta_{\max}$ and let $\widehat{\omega}_{\leq \Delta_{\max}}$ be the positive unital capwise functional furnished by Theorem 5.74A. Let P_t be any positive-time capwise polynomial built from the inverse-flow representatives used in the proof of Source Theorem 5.74, and let $P_t^{\text{bnd}} \in A_+$ be the corresponding RP-safe bounded positive-time representative obtained by the finite-cap insertion package. Then*

$$\omega_{\text{bd}}(\Theta P_t^{\text{bnd}} P_t^{\text{bnd}}) \geq 0$$

for every $t > 0$, and

$$\left| \widehat{\omega}_{\leq \Delta_{\max}}(\Theta P_t P_t) - \omega_{\text{bd}}(\Theta P_t^{\text{bnd}} P_t^{\text{bnd}}) \right| \longrightarrow 0 \quad (t \downarrow 0).$$

Consequently the capwise OS pairing is positive:

$$\liminf_{t \downarrow 0} \widehat{\omega}_{\leq \Delta_{\max}}(\Theta P_t P_t) \geq 0.$$

Equivalently, reflection positivity is read only through bounded positive-time representatives, and the inverse-flow replacement error tends to zero.

Proof. Because $P_t^{\text{bnd}} \in A_+$, reflection positivity of the tuned bounded Wilson base state gives

$$\omega_{\text{bd}}(\Theta P_t^{\text{bnd}} P_t^{\text{bnd}}) \geq 0.$$

Theorem 5.74A packages the same capwise functional together with the RP-through-bounded-positive-time mechanism, so the only comparison needed is the replacement of the flowed probes in P_t by the RP-safe bounded representatives P_t^{bnd} . The finite-cap insertion package gives this replacement error as $o(1)$ in every fixed mixed channel. Passing to the limit yields the stated positivity transfer. $\square$

Theorem B.13 (Source Theorem 5.74: Lane A finite-cap extension of the Lane C flowed state). *Fix an engineering cutoff $\Delta_{\max}$ and let $A_{\text{loc}}^{(\leq \Delta_{\max})}$ be the corresponding finite truncation*

of the sharp-local algebra. Then the inverse-flow approximants define a unique sharp-local continuum state on $A_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ extending the restriction of the already constructed flowed state. Moreover:

- (a) the inverse-flow approximants converge to the centered renormalized local generators in every fixed capwise channel;
- (b) the resulting finite-cap local Schwinger functions satisfy the Osterwalder–Schrader axioms;
- (c) the restriction of the finite-cap sharp-local state to the capwise flowed subalgebra agrees with the Lane C flowed state, and its restriction to the bounded positive-time algebra agrees with the tuned-dyadic bounded Wilson base state.

Proof of Source Theorem 5.74. Fix the engineering cutoff Δ_{max} and abbreviate the capwise sharp-local algebra by $A_{\text{loc}}^{(\leq \Delta_{\text{max}})}$. The theorem-level finite-cap bridge is now explicit: Theorem 5.74A packages, from Source Corollary F.330A, Source Theorem F.331, and Source Corollary 5.70, the unique positive unital capwise functional on the abstract mixed algebra, while Theorem B.12 isolates the finite-cap positivity mechanism itself. Source Theorem 5.74 then adds only the identification of the formal centered symbols with the renormalized local generators, the capwise OS field realization, and the compatibility clauses later split off as Source Corollary 5.75.

For part (a), Theorem 5.74A already gives the capwise mixed moments and their uniqueness. The inverse-flow limits therefore identify the formal centered local symbols with the corresponding centered renormalized local generators in every fixed mixed capwise channel. This is exactly the sharp-local identification claimed in part (a).

For part (b), the theorem-level inputs are Theorem 5.74A and Theorem B.12. Euclidean invariance and permutation symmetry are inherited from the finite-family packaged bounded states, and the positivity mechanism is now isolated theorem-locally: reflection positivity is read only through RP-safe bounded positive-time representatives in A_+ , and the inverse-flow replacement error tends to zero in the mixed channels defining the capwise OS pairing. The common invariant core of bounded flowed positive-time classes supplied earlier in Lane A then realizes the resulting finite-cap local field polynomials as operators on the reconstructed Hilbert space. Thus Source Theorem 5.74 is the capwise OS-structure theorem, but it is not yet the inductive-union theorem.

For part (c), compatibility is recorded directly on the two bounded subalgebras already fixed in Theorem 5.74A. On the capwise flowed subalgebra the new state agrees with the already constructed Lane C flowed state; on the bounded positive-time algebra it agrees with the tuned dyadic bounded Wilson base state because the same bounded packaged moments are used there. The dyadic OS matrix elements are therefore exactly the same ones already recorded for the bounded base state. $\square$

Dependency summary.

Inputs. Theorem 5.74A and Theorem B.12, Source Corollary F.330A, Source Theorem F.331, and Source Corollary 5.70

Imported standard background. none beyond the standard finite-cap OS reconstruction step already triggered by the reflection-positive capwise state furnished through Theorem 5.74A and Theorem B.12

Conclusion at this node. for one fixed engineering cap, the inverse-flow approximants identify the renormalized local generators and produce the capwise sharp-local OS structure together with bounded-state compatibility; the theorem does not yet pass to the inductive union and does not yet assert bounded-base cyclicity

Used later in. Source Corollary 5.75, Source Corollary 5.76, and the QE3 seam theorem Source Theorem F.212

This theorem is the true finite-cap Lane A extension result used by the core paper and by the later endpoint theorems. It does not yet pass to the full inductive union, and it does not yet assert bounded-base cyclicity.

Remark B.14 (Quantifier note for Source Theorem 5.74). Everything in Source Theorem 5.74 is capwise. The engineering cutoff $\Delta_{\max}$ is fixed before the theorem is invoked, and any constants, invariant cores, or graph norms may depend on that cap. The theorem therefore does not assert a quantitative bound uniform as $\Delta_{\max} \rightarrow \infty$, and it does not by itself construct the inductive-union state.

Corollary B.15 (Source Corollary 5.75: Lane A preserves the tuned-dyadic bounded base state). *Let $\omega^{(u_{\text{ren}})}$ be the full sharp-local state later furnished by Source Corollary 5.76. Then*

$$\omega^{(u_{\text{ren}})} \upharpoonright_{A_+} = \omega_{\text{bd}}^{(u_{\text{ren}})}$$

and, for every bounded $F, G \in A_+$ and every dyadic shift $s \in D_{\text{dyad}}$,

$$\omega^{(u_{\text{ren}})}(\Theta F \tau_s G) = \mathcal{K}_{u_{\text{ren}}}(F, G; s).$$

Proof of Source Corollary 5.75. Part (c) of Source Theorem 5.74 gives the stated restriction and dyadic matrix-element identity on every finite engineering cap. Source Corollary 5.76 passes those compatible identities to the inductive union. $\square$

Remark B.16 (Seam note for Source Corollaries 5.75–5.76). Source Corollary 5.75 is the bounded-state restriction compatibility node and nothing more. The finite-cap sharp-local OS structure used before that passage belongs to Source Theorem 5.74(b), while the actual inductive-union passage belongs only to Source Corollary 5.76.

4. Inductive union, bounded-base cyclicity, and the canonical nontriviality witness

The last part of source Section 5 performs the inductive-union upgrade and exports the first bounded-base cyclicity theorem.

Corollary B.17 (Source Corollary 5.76: full sharp-local scope by finite truncations and inductive union). *The finite-truncation sharp-local states are compatible under the natural inclusions of engineering caps. Consequently they define a unique state on the inductive union*

$$A_{\text{loc}} = \varinjlim_{\Delta_{\text{max}} \rightarrow \infty} A_{\text{loc}}^{(\leq \Delta_{\text{max}})},$$

which extends the tuned-dyadic bounded Wilson base state on the positive-time algebra and satisfies the Osterwalder–Schrader axioms on the full sharp-local algebra generated by flowed curvature composites.

Theorem B.18 (Source Theorem 5.77: bounded positive-time base algebra is cyclic in the full sharp-local reconstruction). *Let $(\mathcal{H}_{\text{loc}}, \Omega_{\text{loc}}, H_{\text{loc}})$ be the OS reconstruction of the full sharp-local state. Then:*

- (i) every polynomial in positive-time renormalized local generators is approximated in the reconstructed Hilbert space by bounded positive-time cylinder observables;*
- (ii) the image of the bounded positive-time algebra is dense in $\mathcal{H}_{\text{loc}}$, i.e. Ω_{loc} is cyclic for the bounded base algebra inside the full sharp-local reconstruction;*
- (iii) the closed linear span of the centered bounded classes is the vacuum-orthogonal sector.*

Proof of Source Corollary 5.76. For each engineering cutoff $\Delta_{\max}$, Source Theorem 5.74 constructs a sharp-local state

$$\omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}$$

on the finite truncation $A_{\text{loc}}^{(\leq \Delta_{\max})}$. Fix $\Delta_{\max} \leq \Delta'_{\max}$. We prove compatibility on the smaller cap by reducing to the three generator classes that span it.

Step 1: overlap words in the bounded and flowed sectors. If W is a word built only from bounded positive-time observables and capwise flowed observables lying in $A_{\text{loc}}^{(\leq \Delta_{\max})}$, then Source Theorem 5.74(c) identifies the restrictions of both capwise states with the same tuned bounded base state on A_+ and the same flowed state on the capwise flowed algebra. Hence

$$\omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}(W) = \omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})}(W)$$

for every such overlap word W .

Step 2: overlap words containing renormalized local generators. Let

$$W = Q(B_1, \dots, B_r, O_1, \dots, O_s)$$

be a noncommutative word in which each B_i lies in the overlap of the bounded or flowed sectors and each O_j is a renormalized local generator whose quotient label already occurs below the smaller cap. The coherent normalization scheme fixes those quotient labels before the larger cap is chosen. For each O_j choose the inverse-flow approximants $O_{j,t}$ supplied by Source Theorem 5.74; because the generator already belongs to the smaller cap, the same approximating family $O_{j,t}$ is admissible in both caps. Set

$$W_t := Q(B_1, \dots, B_r, O_{1,t}, \dots, O_{s,t}).$$

Then W_t lies in the overlap algebra generated by bounded and flowed observables, so Step 1 gives

$$\omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})}(W_t) = \omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}(W_t) \quad (t > 0).$$

Source Theorem 5.74(a) identifies each O_j as the mixed-channel limit of $O_{j,t}$ in both caps. Passing to the limit $t \downarrow 0$ therefore yields

$$\omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})}(W) = \omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}(W)$$

for every overlap word W in the smaller cap.

Step 3: state equality on the overlap algebra. Finite linear combinations of overlap words span $A_{\text{loc}}^{(\leq \Delta_{\max})}$. By Steps 1–2 the two capwise states agree on that spanning set, hence

$$\omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})} \upharpoonright_{A_{\text{loc}}^{(\leq \Delta_{\max})}} = \omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}.$$

Thus the finite-cap states are compatible under the natural inclusions.

Now let

$$A_{\text{loc}}^{\text{alg}} := \bigcup_{\Delta_{\max}} A_{\text{loc}}^{(\leq \Delta_{\max})}$$

with the obvious inductive-union algebra structure. Every element of $A_{\text{loc}}^{\text{alg}}$ lies in some finite cap, so define its expectation by evaluating it in any cap containing it. The compatibility proved above makes this independent of the chosen cap, and positivity / unitality are inherited capwise. This yields a well-defined state on the algebraic inductive union

$$A_{\text{loc}} = \varinjlim_{\Delta_{\max} \rightarrow \infty} A_{\text{loc}}^{(\leq \Delta_{\max})}.$$

Finally, every OS test polynomial involves only finitely many generators, hence belongs to one finite cap. For that polynomial, the Euclidean invariance, symmetry, and reflection-positivity clauses are exactly the clauses already proved for the corresponding capwise state in Source Theorem 5.74, so the Osterwalder–Schrader axioms pass verbatim to the inductive union. The restriction to the bounded positive-time algebra is the same tuned-dyadic bounded Wilson base state on every cap and therefore on the full sharp-local algebra. $\square$

Dependency summary.

Inputs. Source Theorem 5.74 together with the explicit overlap comparison on shared generators and inverse-flow representatives fixed by the coherent normalization scheme

Imported standard background. none

Conclusion at this node. the compatible finite-cap sharp-local states glue to one qualitative inductive-union state on the full sharp-local algebra, extending the tuned bounded base state and preserving the OS axioms on finite test polynomials

Used later in. Source Theorem 5.77, Source Theorem F.212, and Source Theorem M.1

Remark B.19 (Quantifier note for Source Corollary 5.76). Source Corollary 5.76 is qualitative glueing only. It identifies compatible finite-cap states on overlaps and defines the inductive-union sharp-local state, but it does not promote the capwise inverse-control, remainder, or graph-norm bounds to a statement uniform over the full union.

Proof of Source Theorem 5.77. Fix $P \in P_{+, \text{sharp}}$ and write

$$P = Q(O_1, \dots, O_r)$$

for a fixed noncommutative polynomial Q in finitely many positive-time renormalized local generators from one engineering cap $\Delta_{\max}$. For each generator O_ν , choose a sequence of RP-safe bounded positive-time representatives

$$B_{\nu, n} \in A_+, \quad n \in \mathbb{N},$$

obtained from the inverse-flow representatives of Source Theorem 5.74 along any fixed sequence $t_n \downarrow 0$, so that

$$\|[B_{\nu, n}] - [O_\nu]\|_{\mathcal{H}_{\text{loc}}} \longrightarrow 0 \quad (n \rightarrow \infty)$$

for each ν . Define the bounded approximating sequence

$$P_n^{\text{bnd}} := Q(B_{1, n}, \dots, B_{r, n}) \in A_+.$$

Because the cap is fixed and Q has finite degree, all factors act on the same finite-cap invariant core, and the remaining bounded factors stay uniformly controlled in the finitely many capwise graph norms relevant to P . Expanding the noncommutative polynomial difference by the telescoping identity gives

$$P_n^{\text{bnd}} - P = \sum_{\nu=1}^r T_{\nu, n} (B_{\nu, n} - O_\nu),$$

where each $T_{\nu, n}$ is a finite linear combination of monomials in the other factors. Hence there is a cap-dependent constant $C_P < \infty$ such that

$$\|[P_n^{\text{bnd}}] - [P]\|_{\mathcal{H}_{\text{loc}}} \leq C_P \sum_{\nu=1}^r \|[B_{\nu, n}] - [O_\nu]\|_{\mathcal{H}_{\text{loc}}}.$$

Since the right-hand side tends to 0, every positive-time sharp-local polynomial class is a norm limit of bounded positive-time cylinder classes. This proves item (i).

To prove density, fix $\psi \in \mathcal{H}_{\text{loc}}$ and $\varepsilon > 0$. Choose a Cauchy sequence of positive-time sharp-local polynomials $P^{(m)} \in P_{+, \text{sharp}}$ whose OS classes converge to ψ , and choose m so large that

$$\|\psi - [P^{(m)}]\|_{\mathcal{H}_{\text{loc}}} < \varepsilon/2.$$

Applying item (i) to the single polynomial $P^{(m)}$, choose n such that

$$\|[P_n^{\text{bnd}}] - [P^{(m)}]\|_{\mathcal{H}_{\text{loc}}} < \varepsilon/2.$$

Then $P_n^{\text{bnd}} \in A_+$ and

$$\|\psi - [P_n^{\text{bnd}}]\|_{\mathcal{H}_{\text{loc}}} < \varepsilon.$$

Because ψ and ε were arbitrary, the image of A_+ is dense in $\mathcal{H}_{\text{loc}}$. This proves item (ii).

Finally, every bounded class decomposes as

$$[F] = [F - \omega(F)] + \omega(F) \Omega_{\text{loc}}.$$

Thus the closed linear span of Ω_{loc} together with the centered bounded classes contains the dense image of A_+ and is therefore all of $\mathcal{H}_{\text{loc}}$. Each centered bounded class is orthogonal to Ω_{loc} , so their closed linear span is contained in $\Omega_{\text{loc}}^\perp$. Conversely, any $\eta \in \Omega_{\text{loc}}^\perp$ may be approximated by bounded classes $[F_n]$ from item (ii), and replacing $[F_n]$ by $[F_n - \omega(F_n)]$ removes only the vacuum component. Hence

$$\overline{\text{span}}\{[F - \omega(F)] : F \in A_+\} = \Omega_{\text{loc}}^\perp.$$

This proves item (iii). □

Dependency summary.

Inputs. Source Theorem 5.74, Source Corollary 5.76, and the RP-safe bounded positive-time representatives constructed on each finite engineering cap

Imported standard background. none

Conclusion at this node. the bounded positive-time base algebra is dense in the reconstructed full sharp-local Hilbert space, and the closed linear span of its centered classes is the vacuum-orthogonal sector; no new spectral-gap statement is proved here

Used later in. Source Corollary F.213 and Source Theorem M.1

Remark B.20 (Finiteness note for Source Theorem 5.77). Source Theorem 5.77 is the first full-union density export on Lane A. In later QE3 uses, every sharp-local polynomial vector is first placed in a finite engineering cap and approximated there by bounded positive-time representatives; only afterward is the full-union density conclusion read off. No full-union familywise graph-norm estimate is claimed here.

Theorem B.21 (Source Theorem 5.78: canonical scalar nontriviality witness). *Let E^{ren} be the centered CP-even dimension-four scalar field picked out by the canonical scalar block, and let $\varphi_{*,S}$ be the source reference test function. Then*

$$\omega^{(u_{\text{ren}})}\left(\Theta E^{\text{ren}}(\varphi_{*,S})E^{\text{ren}}(\varphi_{*,S})\right) = 1.$$

Consequently the corresponding OS class is nonzero, the field is not a multiple of the identity, and the full adjoint-local sharp-local theory is nontrivial in its vacuum sector.

The source concludes by stressing that the inductive-limit step is a compatibility argument rather than a second renormalization problem: once the canonical block basis and normalization moments are fixed, passing to a larger engineering cap preserves the already constructed local fields literally inside the larger cap.

Remark B.22. Within the architecture of the core paper, the role split from source Section 5 must remain literal. Core Theorem 5.9 is the sole mixed-correlator closure node at finite cap; core Theorem 5.10 is the finite-cap sharp-local extension; core Corollary 5.11 is the inductive-union passage; core Theorem 5.13 is the first bounded-base cyclicity export. This appendix keeps those distinctions explicit in native REVTeX form while preserving the theorem-level content of the March 16, 2026 source block.

Appendix C: Native transcription of the support analytic files (source pp. 116–128)

This appendix replaces the former archival proof-page block from source pp. 116–128 with a native REVTeX presentation of the continuum and lattice kernel infrastructure together with the insertion-multiplier support lemmas. In the proof architecture of the companion, these materials are support theorems rather than new theorem endpoints: they supply the Gaussian kernel bounds, the uniform lattice heat-kernel estimates, the discrete parametrix bounds, and the one-step Lipschitz control that feed the Lane A flow-localization and coefficient-transport files.

1. Continuum parametrix and Gaussian kernel bounds

The first support block establishes the continuum Gaussian kernel estimates used in the linearized DeTurck analysis.

Proposition C.1 (Source Proposition B.1: coefficient bounds on the good-flow chart). *There exist constants $A_0, W_0, W_1 \geq 0$ such that for all $\tau \in (0, t_*]$,*

$$\|A_\alpha(\tau, \cdot)\|_{L^\infty(\mathbb{R}^4)} \leq A_0, \quad \|W(\tau, \cdot)\|_{L^\infty(\mathbb{R}^4)} \leq W_0 + W_1\tau^{-1/2}.$$

These bounds are verified on the good-flow chart and are the coefficient input for the continuum parametrix.

Lemma C.2 (Source Lemma B.3: derivative domination by a fatter Gaussian). *For each integer $m \geq 0$ there exists a constant $C_m < \infty$ such that for all $\sigma > 0$ and all $x \in \mathbb{R}^4$,*

$$|\nabla_x^m p(\sigma, x)| \leq C_m \sigma^{-m/2} p(2\sigma, x),$$

where $p(\sigma, x)$ is the four-dimensional heat kernel.

Lemma C.3 (Source Lemma B.4: Volterra inequality for a Gaussian majorant). *Under the coefficient bounds above, the Picard iterates for the continuum kernel satisfy a Volterra-type Gaussian majorant inequality with a universal constant $C^\sharp \geq 1$. This is the basic estimate that feeds the fractional Grönwall argument.*

Lemma C.4 (Source Lemma B.5: fractional Grönwall bound). *If a nonnegative function obeys a Volterra inequality with a $(\tau - \rho)^{-1/2}$ singular kernel and a bounded zeroth-order term, then it admits an exponential majorant of the form required by the continuum parametrix argument.*

Theorem C.5 (Source Theorem B.6: continuum Gaussian upper bound with explicit dependence). *Assume the good-flow coefficient bounds. Then the Picard iteration converges to a fundamental solution $K(t, s; x, y)$ of the continuum linearized system, and there exist universal constants $c \in (0, 1)$ and $C \geq 1$ such that for all $0 < s < t \leq t_*$,*

$$\begin{aligned} \|K(t, s; x, y)\|_{\text{op}} &\leq C(t-s)^{-2} \exp\left(-c \frac{|x-y|^2}{t-s}\right) \\ &\quad \times \exp\left(CA_0^2(t-s) + CW_0(t-s) + 2CW_1(\sqrt{t} - \sqrt{s})\right). \end{aligned}$$

Moreover, for every integer $m \geq 1$ one has the corresponding derivative bound

$$\begin{aligned} \|\nabla_x^m K(t, s; x, y)\|_{\text{op}} &\leq C_m(t-s)^{-(2+m/2)} \exp\left(-c \frac{|x-y|^2}{t-s}\right) \\ &\quad \times \exp\left(CA_0^2(t-s) + CW_0(t-s) + 2CW_1(\sqrt{t} - \sqrt{s})\right). \end{aligned}$$

Proof synopsis. The source proof rewrites the linearized DeTurck equation in divergence form, builds a Picard iteration around the free heat kernel, controls the nonlinear correction by a Gaussian Volterra majorant, and closes the iteration with a fractional Grönwall bound. The theorem-level output is the explicit Gaussian kernel estimate later invoked in the Lane A flow-localization package..

2. Uniform lattice heat-kernel bounds

The second support block translates the continuum Gaussian regime into a uniform-in- a lattice estimate.

Lemma C.6 (Source Lemma C.1: scaling reduction to the unit lattice). *Let $x = ax'$, $y = ay'$ with $x', y' \in \mathbb{Z}^d$ and set $\tau = t/a^2$. Then the lattice heat kernel satisfies*

$$p_t^{(a)}(x, y) = a^{-d} p_\tau^{(1)}(x', y'),$$

and every lattice finite difference of order m scales by an additional factor a^{-m} .

Lemma C.7 (Source Lemma C.2: on-diagonal bound). *There exists $C_d < \infty$ such that for all $t > 0$,*

$$p_t(0) \leq C_d(1+t)^{-d/2}.$$

Lemma C.8 (Source Lemma C.3: tilt bound). *For any $t > 0$, $z \in \mathbb{Z}^d$, and any $\theta \in \mathbb{R}^d$,*

$$|p_t(z)| \leq p_t(0) \exp\left(-\theta \cdot z + 2t \sum_{j=1}^d (\cosh \theta_j - 1)\right).$$

Proposition C.9 (Source Proposition C.4: Gaussian regime for $|z| \leq t$). *There exist constants $c_d, C_d < \infty$ such that for all $t > 0$ and all $z \in \mathbb{Z}^d$ with $|z| \leq t$,*

$$|p_t(z)| \leq C_d t^{-d/2} \exp\left(-c_d \frac{|z|^2}{t}\right).$$

Lemma C.10 (Source Lemma C.5: Fourier multipliers for finite differences). *Every unit-lattice forward-difference operator of order m acts on the Fourier representation of the lattice heat kernel by multiplication with a bounded symbol $M_\alpha(k)$ satisfying a polynomial growth estimate in $|k|$.*

Proposition C.11 (Source Proposition C.6: finite-difference Gaussian regime). *Fix $m_{\max} \geq 0$. Then for every multi-index α with $|\alpha| = m \leq m_{\max}$ there are constants $c_d, C_{d,m}$ such that for all $t > 0$ and all $z \in \mathbb{Z}^d$ with $|z| \leq t$,*

$$|\nabla^\alpha p_t(z)| \leq C_{d,m} t^{-(d+m)/2} \exp\left(-c_d \frac{|z|^2}{t}\right).$$

Theorem C.12 (Source Theorem C.7: uniform lattice heat-kernel bounds in continuum variables). *Fix $d = 4$, a time horizon $t_* > 0$, a radius $R > 0$, and an integer $m_{\max} \geq 0$. Then there exist $a_0 \in (0, 1)$, $c \in (0, 1)$, and constants $C_m < \infty$ for $m = 0, 1, \dots, m_{\max}$ such that for all lattice spacings $a \in (0, a_0]$, all $0 < s < t \leq t_*$ with $t - s \geq a^2$, and all $x, y \in a\mathbb{Z}^4$ with $|x - y| \leq R$,*

$$|p_{t-s}^{(a)}(x, y)| \leq C_0 (t - s)^{-2} \exp\left(-c \frac{|x - y|^2}{t - s}\right),$$

and

$$|(\nabla_x^{(a)})^m p_{t-s}^{(a)}(x, y)| \leq C_m (t - s)^{-(2+m/2)} \exp\left(-c \frac{|x - y|^2}{t - s}\right) \quad (m \leq m_{\max}).$$

The theorem-level use of this block is direct: it supplies the lattice heat-kernel input needed to carry the continuum parametrix argument into the lattice DeTurck setting uniformly in the lattice spacing.

3. Lattice parametrix and one-step multiplier support

The remaining support block packages the discrete Duhamel / parametrix argument and the one-step Lipschitz estimate later used in the insertion multiplier file.

Proposition C.13 (Source Proposition D.1: uniform coefficient bounds in the bounded regime). *Fix $0 < s < t \leq t_*$. Assume there exist constants $B_0, V_0, V_1 \geq 0$ such that for all $r \in [s, t]$,*

$$\max_{1 \leq j \leq 4} \|b_j(r, \cdot)\|_{\ell^\infty(a\mathbb{Z}^4)} \leq B_0, \quad \|V(r, \cdot)\|_{\ell^\infty(a\mathbb{Z}^4)} \leq V_0 + V_1 r^{-1/2}.$$

Lemma C.14 (Source Lemma D.2: existence/uniqueness in the bounded regime). *Under the lattice coefficient bounds above, the discrete Volterra equation for the lattice fundamental solution has a unique kernel solution obtained by Picard iteration; the resulting kernel is absolutely summable in the second variable and defines a bounded operator on ℓ^∞ .*

Theorem C.15 (Source Theorem D.3: lattice Gaussian upper bounds and finite-difference derivatives). *Assume the lattice bounded-regime coefficient bounds. Fix an integer $m \geq 0$. Then there exist constants $C_m = C_m(B_0, V_0, V_1, t_*)$ and $c = c(B_0, t_*) > 0$ such that for all $a \in (0, 1]$, all $0 < s < t \leq t_*$, and all $x, y \in a\mathbb{Z}^4$,*

$$|(\nabla_x^{(a)})^m K_a(t, s; x, y)| \leq C_m(t-s)^{-(2+m/2)} \exp\left(-c \frac{|x-y|^2}{t-s}\right).$$

Lemma C.16 (Source Lemma E.1: one-step Lipschitz bound in (u, K)). *Assume the bulk/insertion seminorm axioms and the insertion integration bound. Then there exist scale-independent constants $C_{\text{Lip},u}$ and $C_{\text{Lip},K}$ such that for every normalized insertion basis element $\Phi_{*,j}$,*

$$\left\| R_j^{\text{ins,good}}(u, K, \Phi_{*,j}) - R_j^{\text{ins,good}}(0, 0, \Phi_{*,j}) \right\|_{j+1} \leq C_{\text{Lip},u} u + C_{\text{Lip},K} \|K\|_j$$

throughout the asymptotically free polydisc.

Proposition C.17 (Source Proposition 5.48: one-step multiplier bounds). *For each sector $* \in \{S, Q, W\}$ there exists a constant $C_{M,*} < \infty$ such that on the asymptotically free ball,*

$$|M_{*,j} - 1| \leq C_{M,*} u_j.$$

After shrinking the small-coupling radius if necessary, one has $|M_{S,j}|, |M_{W,j}| \geq \frac{1}{2}$ throughout the regime of interest.

Proof synopsis. The source proof combines the uniform lattice heat-kernel estimates with the discrete Duhamel formula to obtain a lattice Gaussian parametrix bound, then uses the insertion integration and product estimates to prove a one-step Lipschitz control for normalized insertion basis elements. The multiplier estimate is a finite-dimensional corollary of this one-step Lipschitz bound and of the stable-manifold forcing estimate for the bulk activity..

Remark C.18. For the theorem-level use in the core paper, the present appendix is support material only. It supplies the analytic estimates that make the Lane A flow-localization, insertion transport, and coefficient-multiplier arguments uniform in the dyadic scale and in the lattice spacing; it does not introduce a new endpoint.

Appendix D: Native transcription of the finite-truncation interface and mixed-channel closure file (source pp. 232–280)

This appendix replaces the former Appendix F source-page block with a native REVTeX presentation of the theorem-level finite-truncation interface and mixed-channel closure material. In the architecture of the core paper, the principal outputs here are the canonical finite-truncation SFTE/remainder results, the finite mixed bounded-family preparation node, and the finite-cap mixed-correlator closure theorem. The same source-page range also contains downstream Route 1 reformulation material; to preserve source continuity without disturbing the organization of the detailed arguments, those items are recorded here in a short auxiliary subsection.

1. Finite-cap renormalized local field algebra and intrinsic remainder control

The first part of the source range turns the canonical finite-truncation construction into an explicit finite-cap local-field algebra with a common invariant core.

Theorem D.1 (Source Theorem F.243: dyadic construction of the renormalized local field algebra up to $\Delta_{\max}$). *Assume the weak-coupling curvature-control and weak-window package used in Lane A. Fix $\Delta_{\max} \geq 0$ and the canonical blockwise basis of the exact-dimension quotient formalism. Then there exist:*

- (i) *a graded triangular matrix function $Z^{(\leq \Delta_{\max})}(t)$ on $(0, t_*)$,*
- (ii) *for each test function $\varphi \in C_c^\infty(\mathbb{R}^4)$ constants $C_{\varphi, \Delta_{\max}} < \infty$ and $\eta > 0$,*

such that the dyadic renormalized multiplet obeys a summable increment bound and therefore converges in L^2 as the dyadic flow time tends to zero. The limit defines a renormalized local field multiplet $O_{\text{ren}}^{(\leq \Delta_{\max})}(\varphi)$ on an explicit common dense invariant core of bounded flowed positive-time classes.

Lemma D.2 (Source Lemma F.244: explicit finite-cap common invariant core). *Fix $\Delta_{\max} \geq 0$ and let $\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ be the span of bounded flowed positive-time classes in the finite-cap OS Hilbert space. Then:*

- (i) *$\mathcal{D}_{\text{flow}}^{(\leq \Delta_{\max})}$ is dense and invariant under the bounded dyadic approximants;*

(ii) for each canonical field component and each bounded flowed positive-time factor, the dyadic approximants form a Cauchy sequence in the OS Hilbert norm and in every fixed finite graph norm relevant to the polynomial field algebra;

(iii) consequently the renormalized field components are closable, preserve the common invariant core, and define a well-posed finite polynomial field algebra there.

Corollary D.3 (Source Corollary F.245: canonical finite-truncation SFTE and intrinsic remainder control). *Fix $\Delta_{\max} \geq 0$ and the canonical blockwise basis. Then the canonical flowed lifts and the renormalized local fields produced by Theorem F.243 satisfy the Lane A small-flow-time expansion on that finite truncation,*

$$Y_t^{(\leq \Delta_{\max})}(f) = C^{(\leq \Delta_{\max})}(t; \mu) O_{\text{ren}}^{(\leq \Delta_{\max})}(f; \mu) + R_t^{(\leq \Delta_{\max})}(f; \mu),$$

with a block-lower-triangular coefficient matrix and a remainder whose matrix elements against every fixed finite insertion family decay after multiplication by the same finite-cap inverse-control function.

Proof synopsis. The source proof packages the canonical one-shell transport, imposes the graded triangular renormalization conditions recursively in the grading order, proves summability of the dyadic shell increments, and upgrades the resulting L^2 control to graph-norm control by the insertion-to-OS transfer theorem. The corollary is the exact finite-cap SFTE/remainder package later consumed by the mixed-channel closure theorem..

2. Auxiliary downstream Route 1 reformulation retained from the same source range

The source-page range 276–279 also contains a downstream Route 1 reformulation block. These results are not part of the core detailed-proof references for Companion II, but they are retained here in native form to preserve continuity with the March 16, 2026 source file.

Theorem D.4 (Source Theorem F.326: secondary Route 1 dense-core bridge via QE3 and OS spectral calculus). *Let $\omega^{(u_{\text{ren}})}$ be the sharp-local continuum state furnished by the Lane A closure chain and let $(\mathcal{H}, \Omega, H)$ be its OS reconstruction. Then the dense positive-time core generated by bounded local cylinder observables is OS-dense and satisfies Euclidean exponential clustering with the same gap rate m_* already obtained on the primary Route 1 spine.*

Theorem D.5 (Source Theorem F.327: identification of the Route 1 bounded state and the Lane A sharp-local state). *For the tuned dyadic sequence attached to u_{ren} , the bounded positive-time state and its dyadic OS matrix-element limits agree with the restriction of the full sharp-local continuum state to the positive-time algebra. Consequently every Route 1 estimate first proved for the bounded dyadic OS matrix elements transfers to the same sharp-local state used in the main theorem.*

Corollary D.6 (Source Corollary F.329: noncollapse of the physical mass scale from one-shot entrance). *Along the tuned ultraviolet sequence associated with u_{ren} , there exists a uniform strictly positive lower bound on the physical mass scale. In the source architecture this is the theorem-level noncollapse consequence of the one-shot entrance theorem together with the exact entrance-scale covariance transfer.*

Remark D.7. These Route 1 reformulation items are included here only because they occupy the same source-page block as Corollary F.330A and Theorem F.331. The stable detailed-proof reference for the core paper's Route 1 theorems remains Companion I.

3. Finite mixed bounded-family packaging and the sole mixed-correlator closure node

The last part of the source range contains the actual finite-cap mixed-channel seam.

Corollary D.8 (Source Corollary F.330A: finite mixed bounded-family packaging on a finite truncation). *Fix $u_{\text{ren}} \in (0, u_*]$ and $\Delta_{\text{max}} \geq 0$, and let*

$$B_{\text{cap}}^{(\leq \Delta_{\text{max}})} := * \left(A_{\text{flow}}^{(\leq \Delta_{\text{max}})}, A_+ \right)$$

be the unital bounded capwise algebra generated by the capwise flowed subalgebra and the bounded positive-time algebra. For every finite family $W \subset B_{\text{cap}}^{(\leq \Delta_{\text{max}})}$, the tuned Wilson expectations of the words generated by W converge jointly along the dyadic trajectory, and therefore define a unique packaged state

$$\omega_{W, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$$

on the generated finite algebra. These packaged states are positive, normalized, compatible under enlargement of the family W , and inherit Euclidean translation covariance on translated finite families.

Theorem D.9 (Source Theorem F.331: finite-cap mixed-correlator extension from finite-family packaged bounded capwise states). *Fix $u_{\text{ren}} \in (0, u_*]$ and $\Delta_{\text{max}} \geq 0$. Let*

$$B_{\text{cap}}^{(\leq \Delta_{\text{max}})} := * \left(A_{\text{flow}}^{(\leq \Delta_{\text{max}})}, A_+ \right)$$

and let $\hat{A}_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ be the unital $$ -algebra generated by $B_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ and the formal centered capwise local symbols. Then, after first fixing a finite family $W \subset B_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ large enough to contain every bounded word appearing in the formula under consideration, the packaged state from Corollary F.330A extends to a mixed-correlator functional on the algebra generated by those words and the centered local symbols, with the following theorem-level consequences:*

- (i) replacing the designated local slot by its bounded inverse-flow representative changes every mixed correlator by $o(1)$ as the flow parameters tend to zero;*
- (ii) the resulting value is independent of the admissible finite family W chosen to package the bounded capwise state;*
- (iii) the positive-time mixed-channel matrix elements are compatible with the same bounded positive-time state that later enters the Lane A finite-cap sharp-local extension theorem.*

This theorem is the first mixed-correlator closure node on the Lane A spine and the sole theorem-level mixed-channel closure node cited by the core paper.

Proof synopsis. The source proof first packages every finite bounded word family into a positive compatible capwise state, then proves that the inverse-flow approximants are Cauchy in every fixed mixed channel against those packaged states. Because compatibility under enlargement of the finite family is built into Corollary F.330A, the limiting mixed-correlator functional is independent of the chosen auxiliary bounded family. This is precisely what turns the finite-cap seam into a theorem-level closure node rather than a diffuse bookkeeping argument..

Remark D.10. Within the architecture of the core paper, the role split must remain literal: Corollary F.330A supplies the finite mixed bounded-family packaging node, while Theorem F.331 is the sole mixed-correlator closure theorem at finite cap. The theorem is fixed-cap and fixed-family in that precise sense: the auxiliary bounded family W is frozen before the closure statement is read, and no inductive-union or full-union quantitative uniformity is asserted here. The inductive-union passage and bounded-base cyclicity occur only later in the Lane A chain.

Appendix E: Native transcription of the normalization cross-check and interface file (source pp. 300–307)

This appendix replaces the former normalization / interface block with a native REVTeX presentation of the source conventions linking the bi-invariant metric, the Casimir and Laplace normalization, the one-loop coefficient bookkeeping, and the canonical exact-dimension quotient machinery. Its theorem-level role in the present companion is to house the exact-dimension quotient identity and coefficient-extraction theorems (source Theorems L.8 and L.11) together with the Lane A interface note distinguishing the finite-truncation file, the mixed-correlator closure node, the inductive-union passage, and bounded-base cyclicity.

1. Threshold closure note and metric/Casimir normalization

The source-page range opens with the threshold closure summary and then records the normalization dictionary used throughout the Lane A coefficient file.

Remark E.1 (Source K.3 / Remark K.1: closure statement in threshold form). Assume the master threshold $\beta \geq \beta_{\text{master}}(G, \rho)$. Then the source architecture closes simultaneously the Lane C unique continuum flow-bundle state, the Lane A local gauge-invariant sharp-local OS theory, and the primary Route 1 mass-gap chain. No additional tuning parameters occur beyond the named threshold constants; any attempted enlargement of the parameter list constitutes a scope change and must be stated as a separate theorem.

Lemma E.2 (Source Lemma L.1: Laplacian/Casimir consistency). *For the fixed bi-invariant metric, the Laplace–Beltrami operator is*

$$\Delta = \sum_{a=1}^{d_G} \widetilde{X}_a^2,$$

and for every irreducible unitary representation π one has

$$\Delta \chi_\pi = -C_2(\pi) \chi_\pi.$$

This is the Casimir sign convention used throughout the manuscript.

Definition E.3 (Source Definition L.2: Dynkin index relative to the fixed metric). Let σ be a finite-dimensional unitary representation of G with differential σ_* . The Dynkin index I_σ

relative to the fixed inner product is defined by

$$-\mathrm{Tr}_\sigma(\sigma_*(X)\sigma_*(Y)) = I_\sigma\langle X, Y\rangle.$$

Lemma E.4 (Source Lemma L.3: quadratic coefficient at the identity). *Let ρ be the fixed Wilson representation. As $X \rightarrow 0$ in the Lie algebra,*

$$1 - \frac{1}{\dim \rho} \Re \mathrm{Tr}_\rho(e^X) = \frac{I_\rho}{2 \dim \rho} \|X\|^2 + O(\|X\|^3).$$

Consequently the Wilson potential has Hessian

$$\nabla^2 V_{\beta, \alpha}(0) = \alpha \beta \frac{I_\rho}{\dim \rho} I.$$

Lemma E.5 (Source Lemma L.4: matching to the conventional one-loop coefficient). *If the gradient-flow coupling satisfies $u = g^2 + O(g^4)$ as $g \rightarrow 0$ and the reference coupling has beta function*

$$\mu \frac{dg}{d\mu} = -\frac{b_0^{\mathrm{conv}}}{16\pi^2} g^3 + O(g^5),$$

then

$$\beta_{\mathrm{GF}}(u) = \mu \frac{du}{d\mu} = -2 \frac{b_0^{\mathrm{conv}}}{16\pi^2} u^2 + O(u^3).$$

Thus the coefficient b_0 used in the Lane C step-scaling lane matches the conventional one-loop coefficient after tree-level normalization; the load-bearing use remains only the strict positivity $b_0 > 0$ proved elsewhere.

2. Component 5–7 interface map and canonical exact-dimension quotient blocks

The source then records the Lane A interface and the canonical exact-dimension block protocol used later in the finite-truncation inverse theorem. Because these two theorems are load-bearing for the finite-cap closure route, we record the proof steps explicitly rather than only by synopsis.

Remark E.6 (Source Appendix L.5: Component 5–7 interface map). For a first reading of Components 5–7, the theorem-level chain is:

graded identity / block coefficients $\implies$ one-shell transport + finite-truncation inverse control $\implies$ finite-cap flowed-sector bridge $\implies$ Lane A closure $\implies$ bounded-base cyclicity.

The interface map records explicitly that Theorem F.331 is the first mixed-correlator closure node, that the promoted theorem-level bridge into Source Theorem 5.74 records the positive/unital finite-cap state input, that Source Theorem 5.74(b) is only the finite-cap sharp-local OS-structure theorem, that Source Corollary 5.75 is only the bounded-state restriction compatibility node, that the inductive-union extension occurs only at Source Corollary 5.76, and that bounded-base cyclicity appears only at Source Theorem 5.77.

Definition E.7 (Source Definition L.6: canonical block norm). Fix an exact-dimension symmetry block with canonical basis $\{e_1, \dots, e_m\}$. For $v = \sum_{i=1}^m c_i e_i$ define

$$\|v\|_{\text{can}} := \max_{1 \leq i \leq m} |c_i|.$$

The corresponding operator norm on linear maps between canonical blocks is denoted $\|\cdot\|_{\text{op,can}}$.

Lemma E.8 (Source Lemma L.7: small-flow substitution respects the exact-dimension quotient). *Let Γ_t denote the algebra map sending each unflowed jet to its flowed counterpart. Then for every homogeneous representative P of exact engineering dimension Δ ,*

$$\Gamma_t P = P + \sum_{m \geq 1} t^m P_{\Delta+2m} + R_{<\Delta},$$

where each $P_{\Delta+2m}$ has exact engineering dimension $\Delta + 2m$ and $R_{<\Delta}$ is lower-dimensional modulo the homogeneous quotient relations. Consequently Γ_t descends to a well-defined map on the exact-dimension associated-graded quotient.

Theorem E.9 (Source Theorem L.8: associated-graded identity on exact-dimension symmetry blocks). *For every exact-dimension symmetry block $V_{\Delta,\epsilon,\lambda}$, the descended small-flow map is the identity after the canonical identification of the unflowed and flowed exact-dimension quotient blocks:*

$$\text{gr}_{\Delta}(\Gamma_t)|_{V_{\Delta,\epsilon,\lambda}} = I_{V_{\Delta,\epsilon,\lambda}}.$$

Equivalently, the tree-level small-flow matching on the exact-dimension quotient block is basis-free and exactly the identity.

Lemma E.10 (Source Lemma L.9: coefficient extraction is well defined on arbitrary exact-dimension quotient blocks). *Coefficient extraction on a canonical exact-dimension symmetry block depends only on the quotient class, both before and after the small-flow substitution.*

Under the canonical identification of the unflowed and flowed exact-dimension blocks, the extracted coefficient vector is unchanged by passing to the exact-dimension associated-graded projection.

Lemma E.11 (Source Lemma L.10: canonical block data are cap-compatible). *The canonical local basis, canonical flowed basis, and canonical block norm are literally the same whether a fixed exact-dimension symmetry block is viewed inside one engineering cap or any larger one. In particular, operator bounds recorded in the canonical block norm are statements about the block itself and not about the ambient truncation.*

Theorem E.12 (Source Theorem L.11: coefficient extraction theorem for arbitrary exact-dimension symmetry blocks). *Fix an exact-dimension symmetry block and its canonical local and flowed bases. Let*

$$C_{\Delta, \epsilon, \lambda}^{\text{UV, tree}}$$

be the tree-level exact-dimension diagonal block of the UV matching matrix obtained by coefficient extraction on the canonical basis classes. Then

$$C_{\Delta, \epsilon, \lambda}^{\text{UV, tree}} = I.$$

In particular, the exact-dimension diagonal UV matching problem on an arbitrary block is an intrinsic finite-dimensional linear-algebra problem attached to the quotient block itself, and the same matrix is seen in every engineering cap containing that block.

Lemma E.13 (Source Lemma L.12: reference scalar normalization for the nontriviality witness). *For the canonical CP-even scalar block, the coherent Lane A normalization may be fixed so that the chosen positive-time reference test function satisfies unit OS norm at every dyadic scale and in every engineering cap containing the scalar block. This is the normalization later used in the canonical scalar nontriviality witness.*

Proof of Source Theorem L.8, Lemma L.9, Lemma L.10, and Source Theorem L.11. For Theorem L.8, start from Lemma L.7: for any homogeneous representative P of exact engineering dimension Δ ,

$$\Gamma_t P = P + \sum_{m \geq 1} t^m P_{\Delta+2m} + R_{<\Delta},$$

with each $P_{\Delta+2m}$ of exact dimension $\Delta + 2m$ and $R_{<\Delta}$ lower-dimensional modulo the homogeneous quotient relations. Passing to the exact-dimension associated-graded quotient discards the higher-dimension terms and kills $R_{<\Delta}$, so the descended map on the exact-dimension quotient sends the class of P to itself. Because the (ϵ, λ) -symmetry decomposition is homogeneous, the same conclusion holds after restriction to any fixed symmetry block $V_{\Delta, \epsilon, \lambda}$, which proves

$$\text{gr}_{\Delta}(\Gamma_t) \Big|_{V_{\Delta, \epsilon, \lambda}} = I_{V_{\Delta, \epsilon, \lambda}}.$$

Lemma L.9 is then immediate from the definition of the exact-dimension quotient. If $P' \equiv P$ modulo lower-dimensional terms and homogeneous quotient relations, then coefficient extraction on the canonical block depends only on the quotient class before the flow substitution. After the substitution, Lemma L.7 shows that lower-dimensional pieces remain lower-dimensional and the relation space is preserved, so coefficient extraction on the flowed exact-dimension block is likewise representative-independent. Under the canonical identification of the unflowed and flowed exact-dimension blocks by the same complete-contraction formulas, the extracted coefficient vector is unchanged.

Lemma L.10 is a compatibility statement. The canonical local basis is fixed block-by-block before any truncation is chosen; the canonical flowed basis is obtained from the same complete-contraction formulas by replacing each curvature or derivative jet by its flowed counterpart; and the coherent renormalization scheme preserves already normalized blocks under cap enlargement. Hence the canonical bases and the canonical block norm are literally the same in every larger engineering cap.

For Theorem L.11, apply coefficient extraction to each canonical basis vector of the exact-dimension symmetry block. By Theorem L.8 and Lemma L.9, the small-flow image of the q th basis vector has exactly the same exact-dimension coefficient vector as the basis vector itself, namely the q th standard basis vector. Therefore the tree-level exact-dimension diagonal UV matching matrix has columns equal to the standard basis vectors and is therefore the identity:

$$C_{\Delta, \epsilon, \lambda}^{\text{UV, tree}} = I.$$

By Lemma L.10, the same canonical local basis, canonical flowed basis, and canonical norm are seen in every larger engineering cap, so the same matrix is obtained whenever the block is embedded in a larger truncation. $\square$

Remark E.14. Within the architecture of the companion, this appendix contains the detailed proof of the exact-dimension quotient results cited by the body and by the core paper. It is not itself the finite-cap closure theorem; rather, it supplies the canonical block data consumed later by the one-shell transport, finite-truncation inverse, and mixed-correlator closure files.

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